Assessment of supply chain of construction materials

Assessment objectives

PRACTICAL ACTION
Technology challenging poverty

Overall objective : Support Government of Nepal/NRA to develop/adopt appropriate strategy for supply chain management of construction materials for reconstruction

Specific objectives;

- Understand the existing supply chain situation : ***Demand supply gap , Quality, Price and Timely Supply***
- Identify the major issues : Policy level and Operational Level
- Propose the appropriate strategy /measures to address the constraints

Methodologies

- ✓ Clusters and Sub- Clusters division (***based on Market centres and accessibility***)
- ✓ Focus Group Discussions (FGD) with earthquake victims at each sub-clusters
- ✓ Key Informant Interview (KII)- ***VDC secretaries, Vendors, Manufacturers, Producers, Transporters***)
- ✓ Interaction and discussion with the stakeholders (***NRA , DDCs, FNCCI, Forum***)
- ✓ District level workshop – Coordinated by FNCCI
- ✓ Previous Studies- (***PDNA, Study Carried out by IOM)***

Limitations

- The field assessment was carried out at **Nuwakot and Rasuwa** districts only. The **national level inferences** are drawn assuming that the findings hold true for other districts as well
- Of the proposed 17 designs, **SMM 1.1, SMC 2.1, BMM 1.1, BMC 2.1** were considered for the materials calculation. All the houses except for BMC 2.1 in accessible area are considered single story. BMC 2.1 in Urban area are considered the double story
- Only the demand for the **individual houses** have been considered in the study (**Institutional building** and **community infrastructures** are not considered)

Findings

Findings are grouped into 4 dimensions of supply chain management

A. Demand and supply Gap

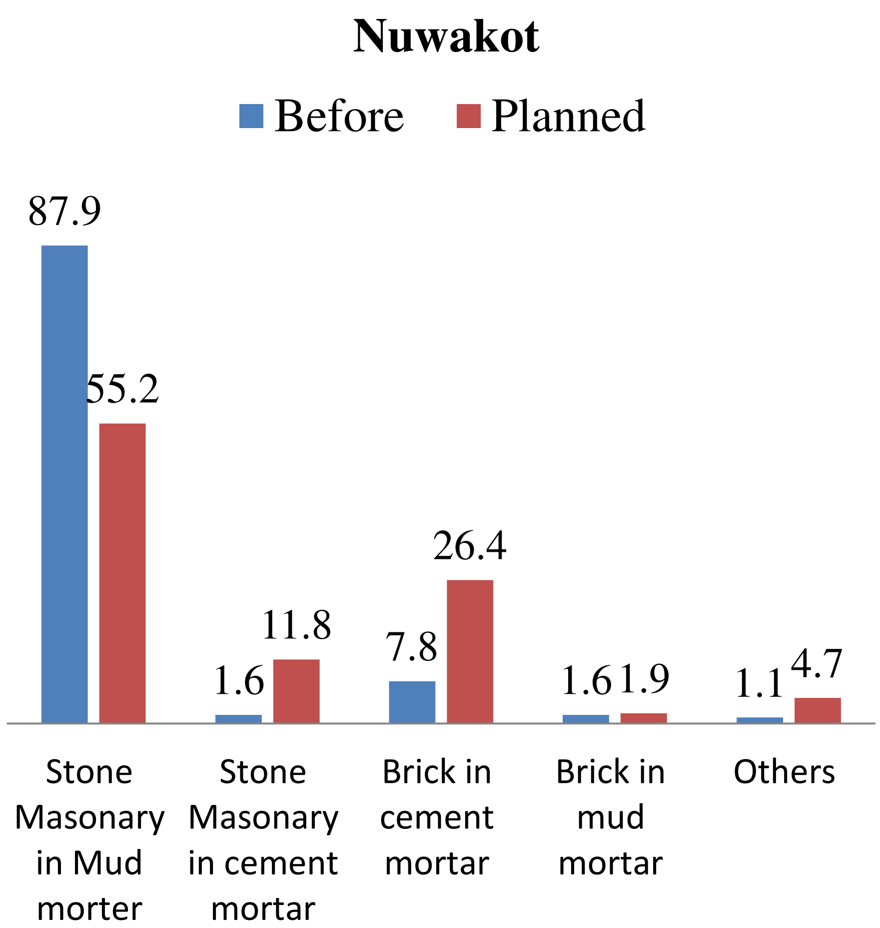
B. Quality

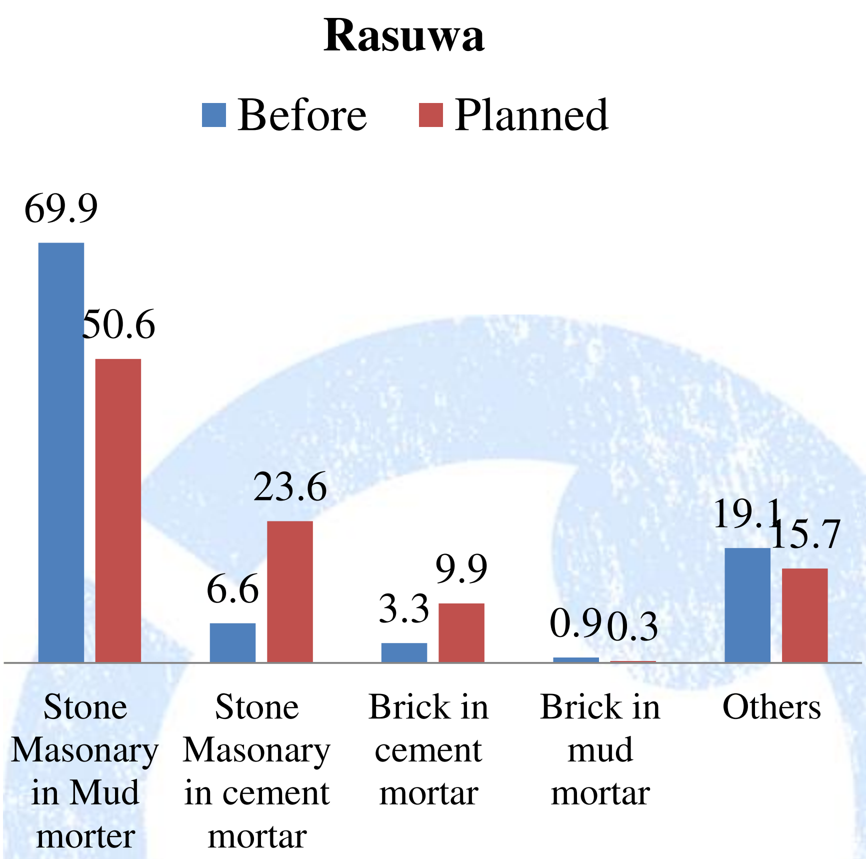
C. Price

D. Timely delivery

A. Demand and supply Gap

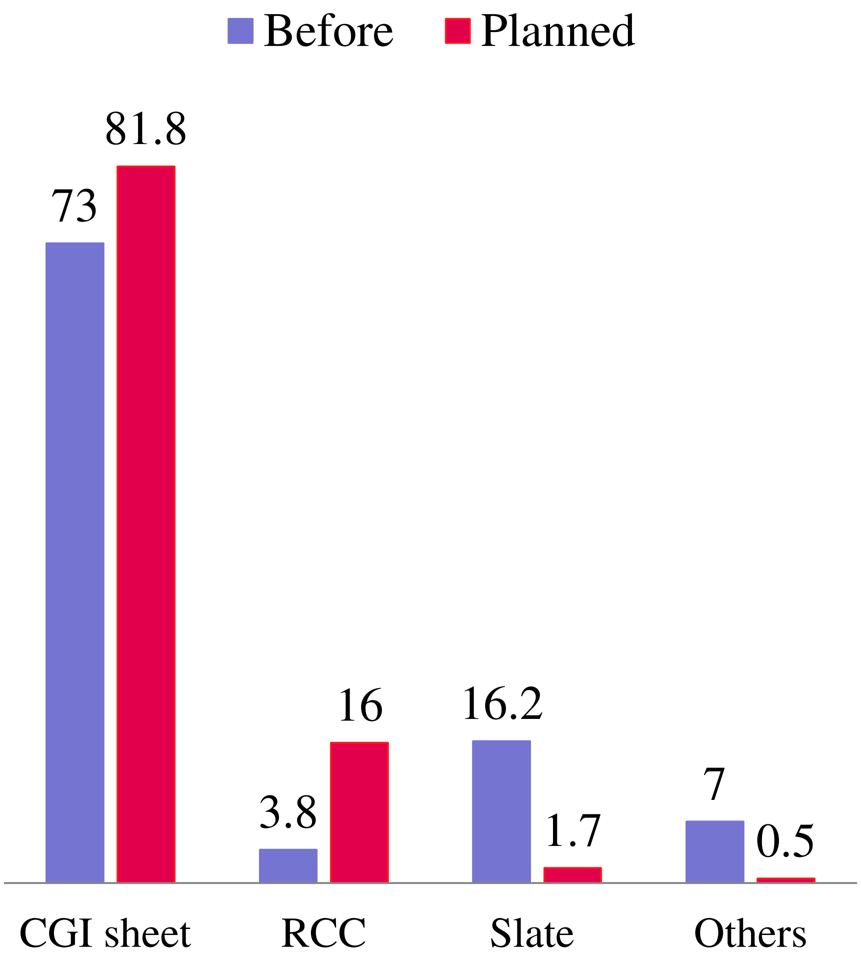
Building Type

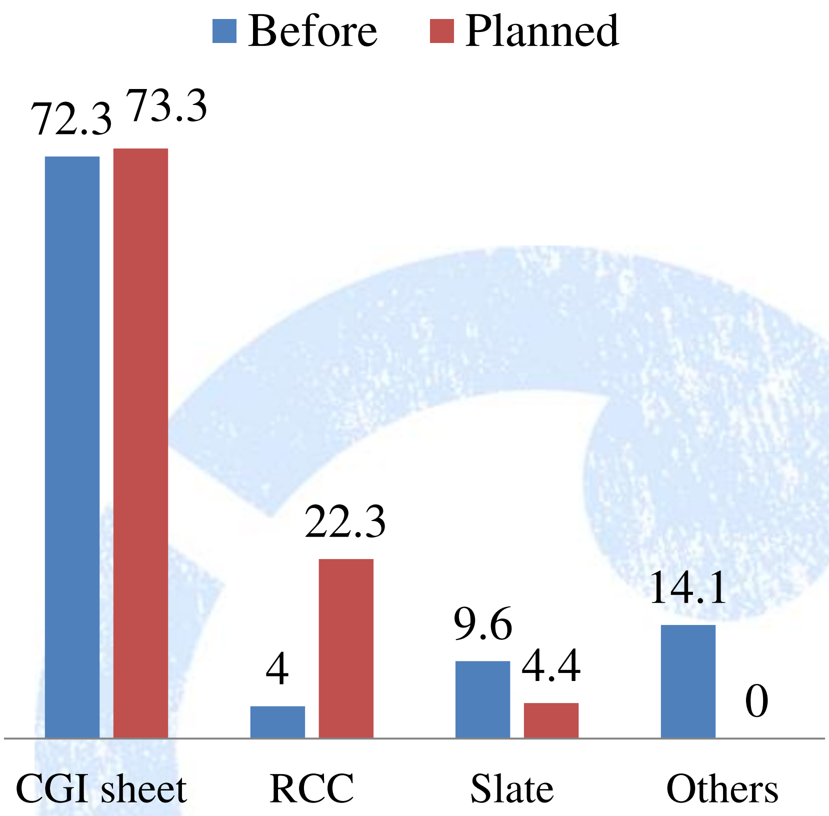




A. Demand and supply Gap

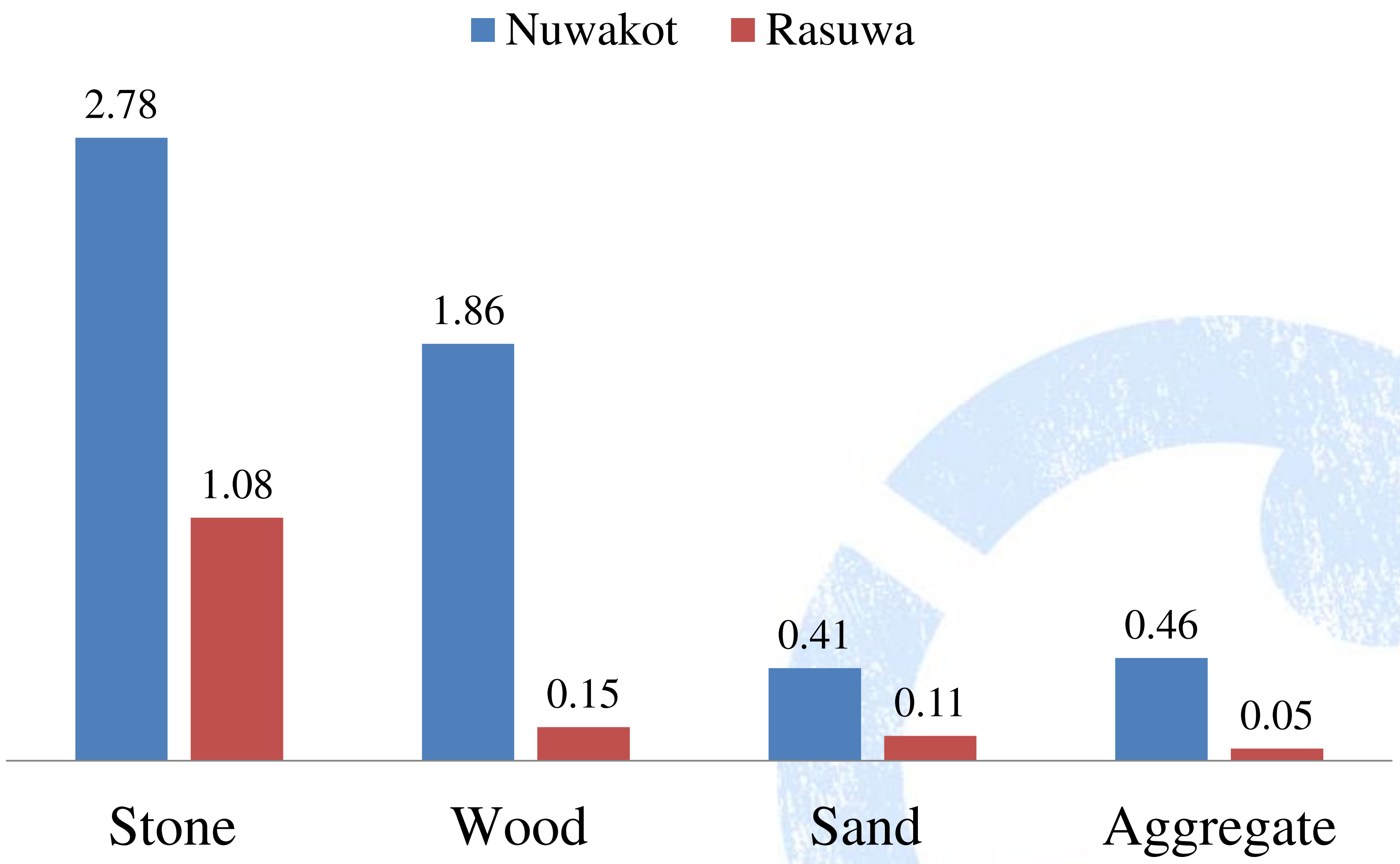
Roof Type



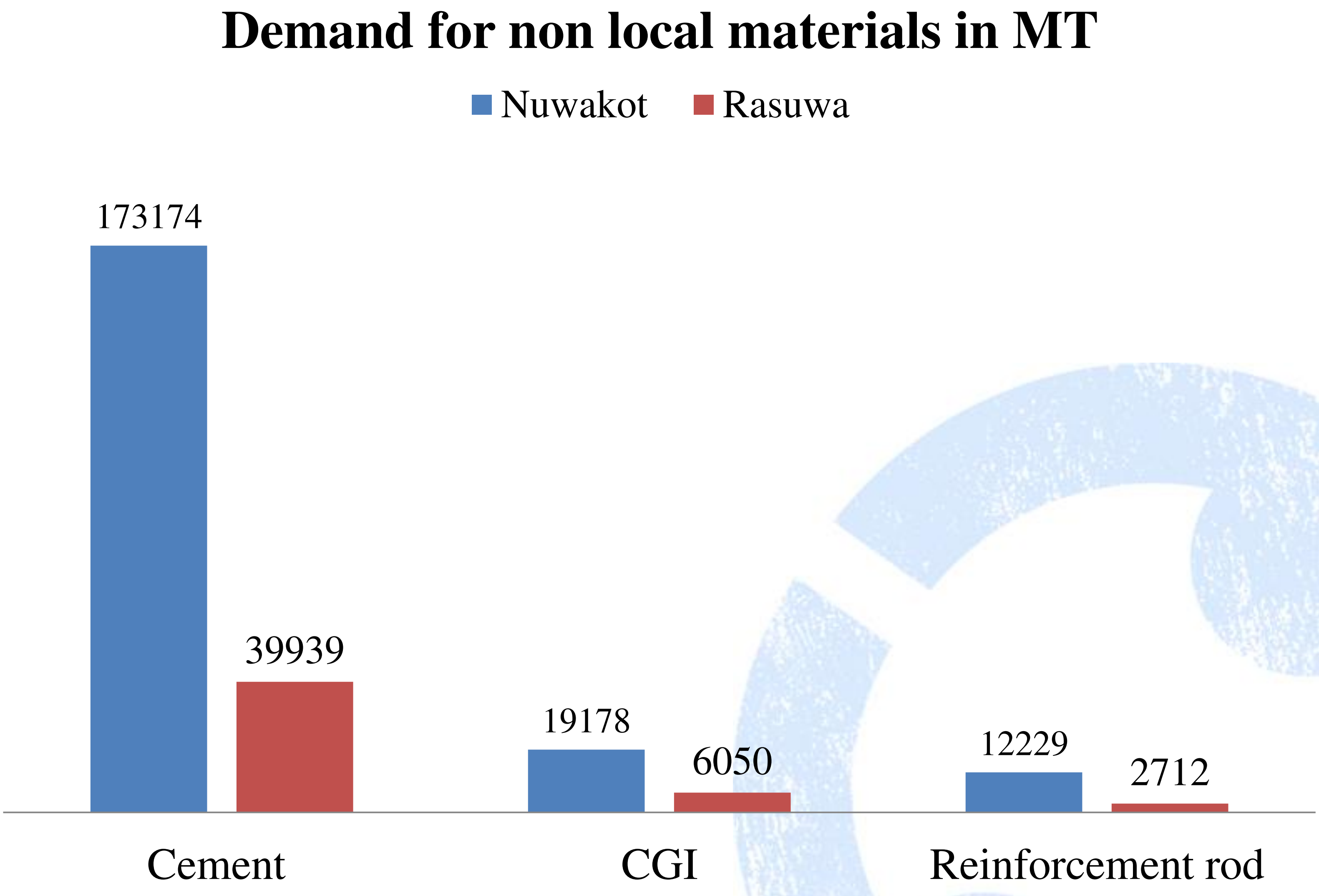


A. Demand and supply Gap

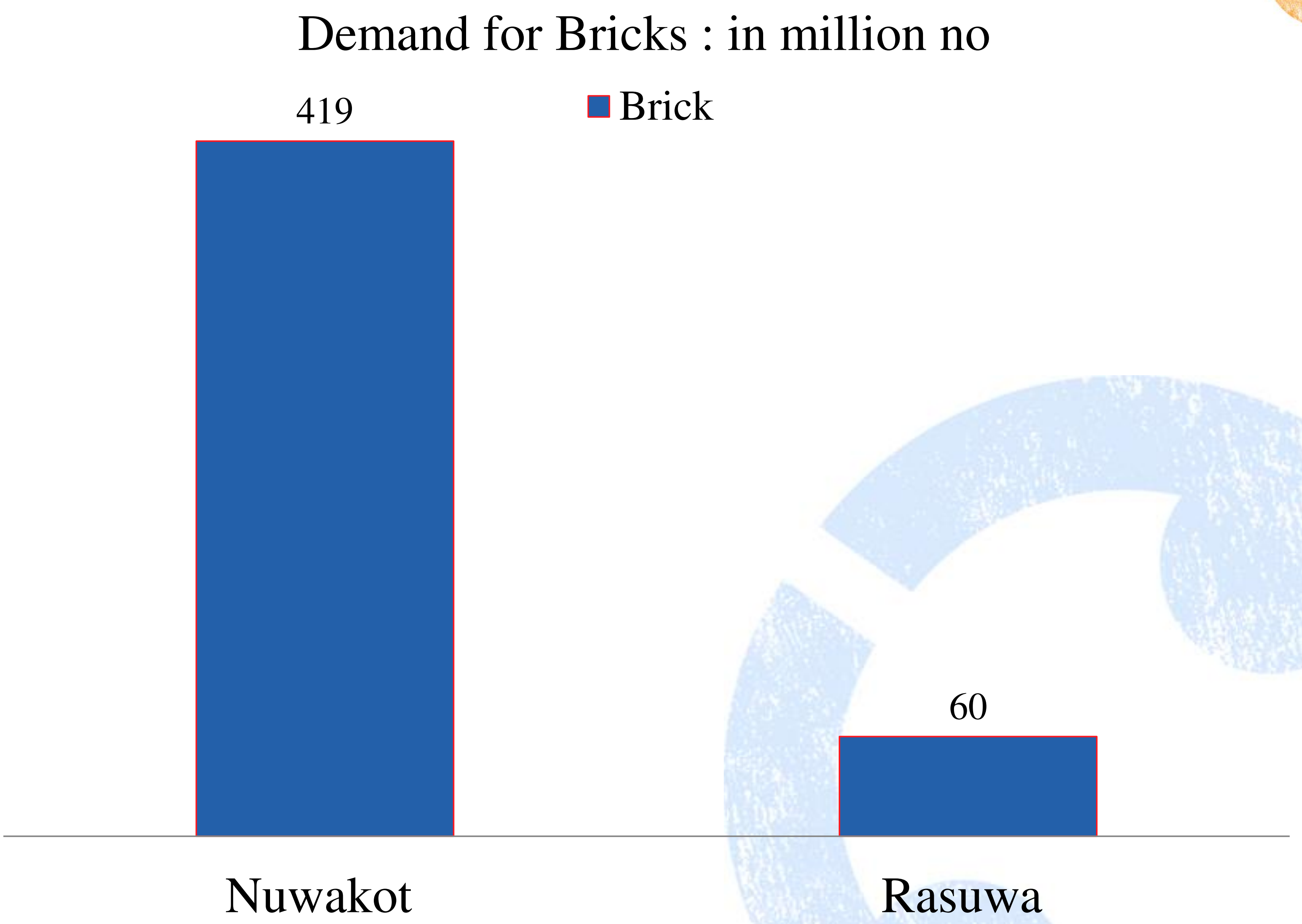
Demand for local Materials in Million Metre cube



A. Demand and supply Gap

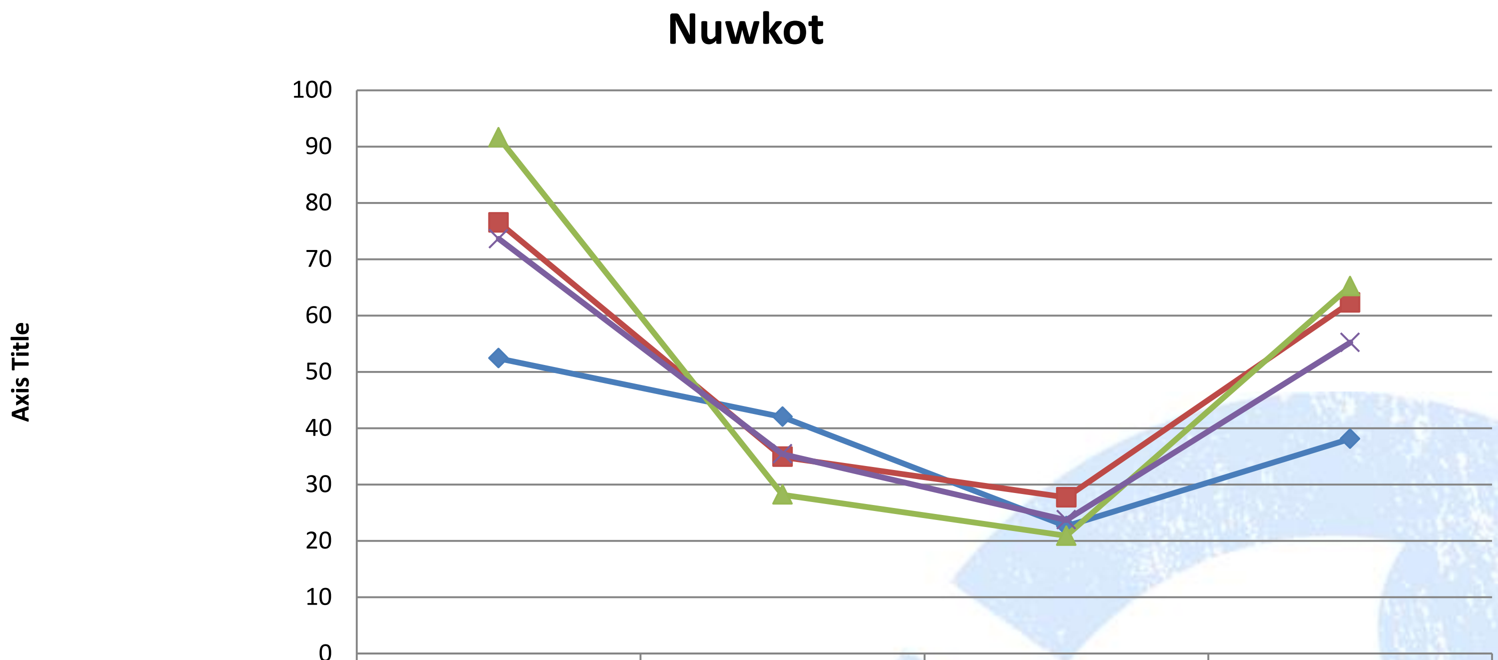


A. Demand and supply Gap



A. Demand and supply Gap

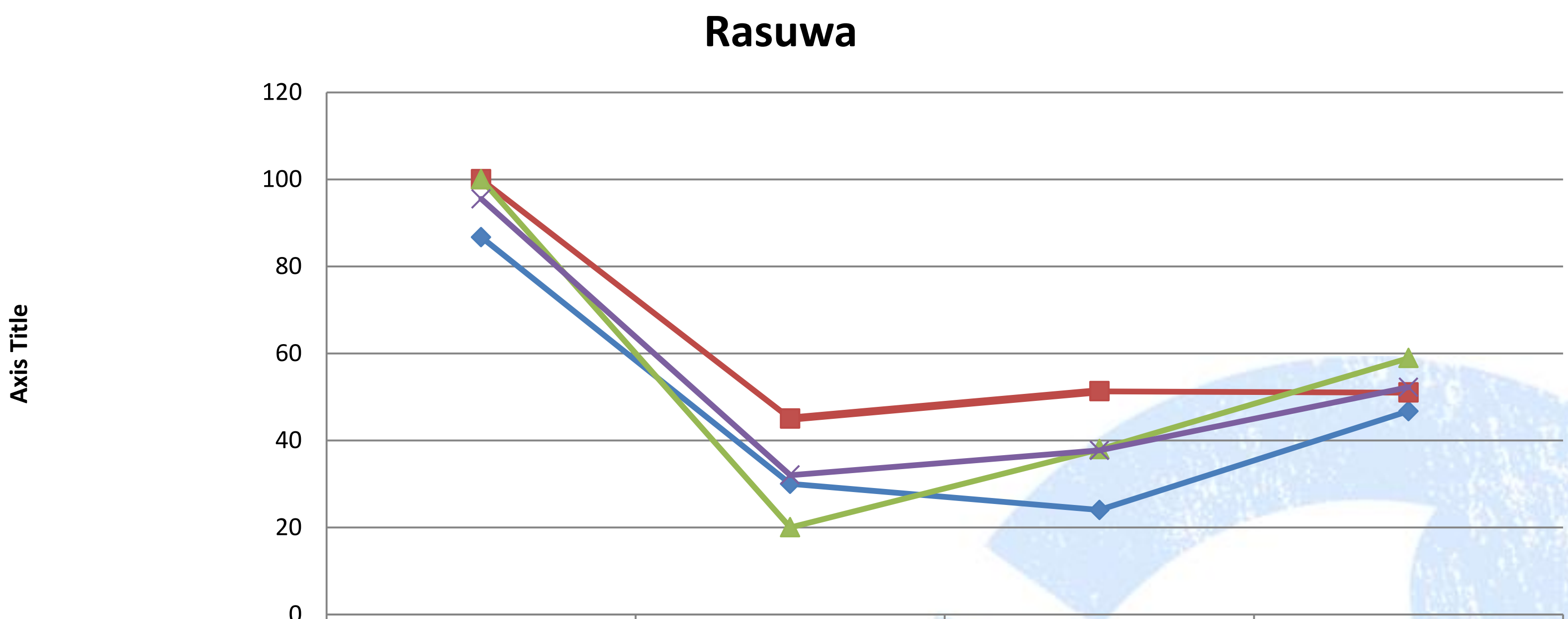
Demand that can be met locally (%)



	Stone	Sand	Aggregate	Wood
Urban	52.4	42	22.6	38.1
Semi Urban	76.5	34.9	27.7	62.3
Rural	91.6	28.2	20.9	65.2
Overall	73.6	35.4	23.7	55.2

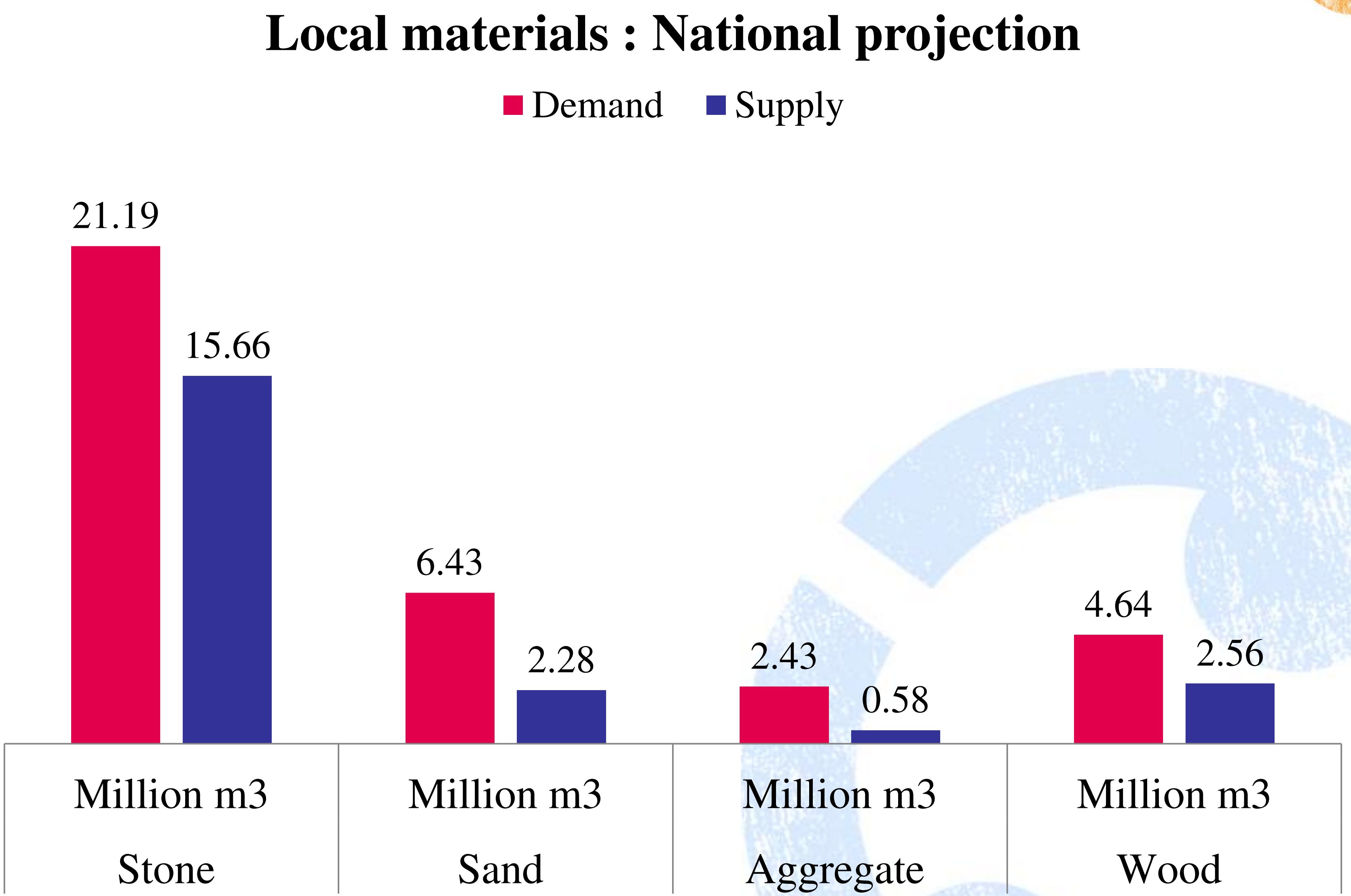
A. Demand and supply Gap

Demand that can be met locally (%)



	Stone	Sand	Aggregate	Wood
Urban	86.7	30	24	46.7
Semi Urban	100	45	51.28	51
Rural	100	20	37.96	58.9
Overall	95.5	32	37.7	52.2

A. Demand and supply Gap



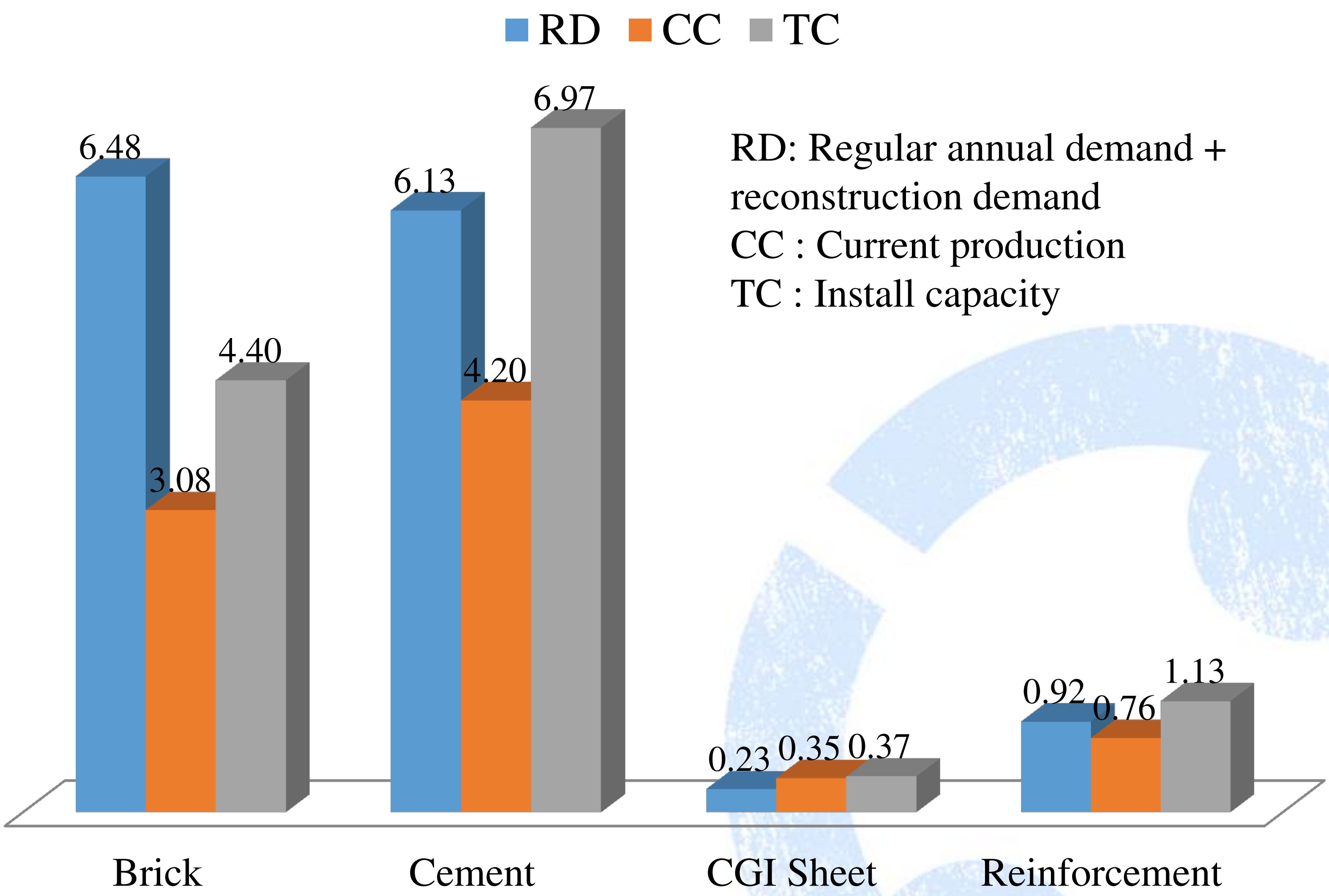
A. Demand and supply Gap

Projected Reconstruction demand for the materials

Construction materials	Unit	Total Reconstruction Demand	Regular annual Demand
Brick	Billion no	6.60	3.2
Cement	Million MT	2.26	5.0
CGI Sheet	Million MT	0.15	0.2
Reinforcement	Million MT	0.14	0.9

A. Demand and supply Gap

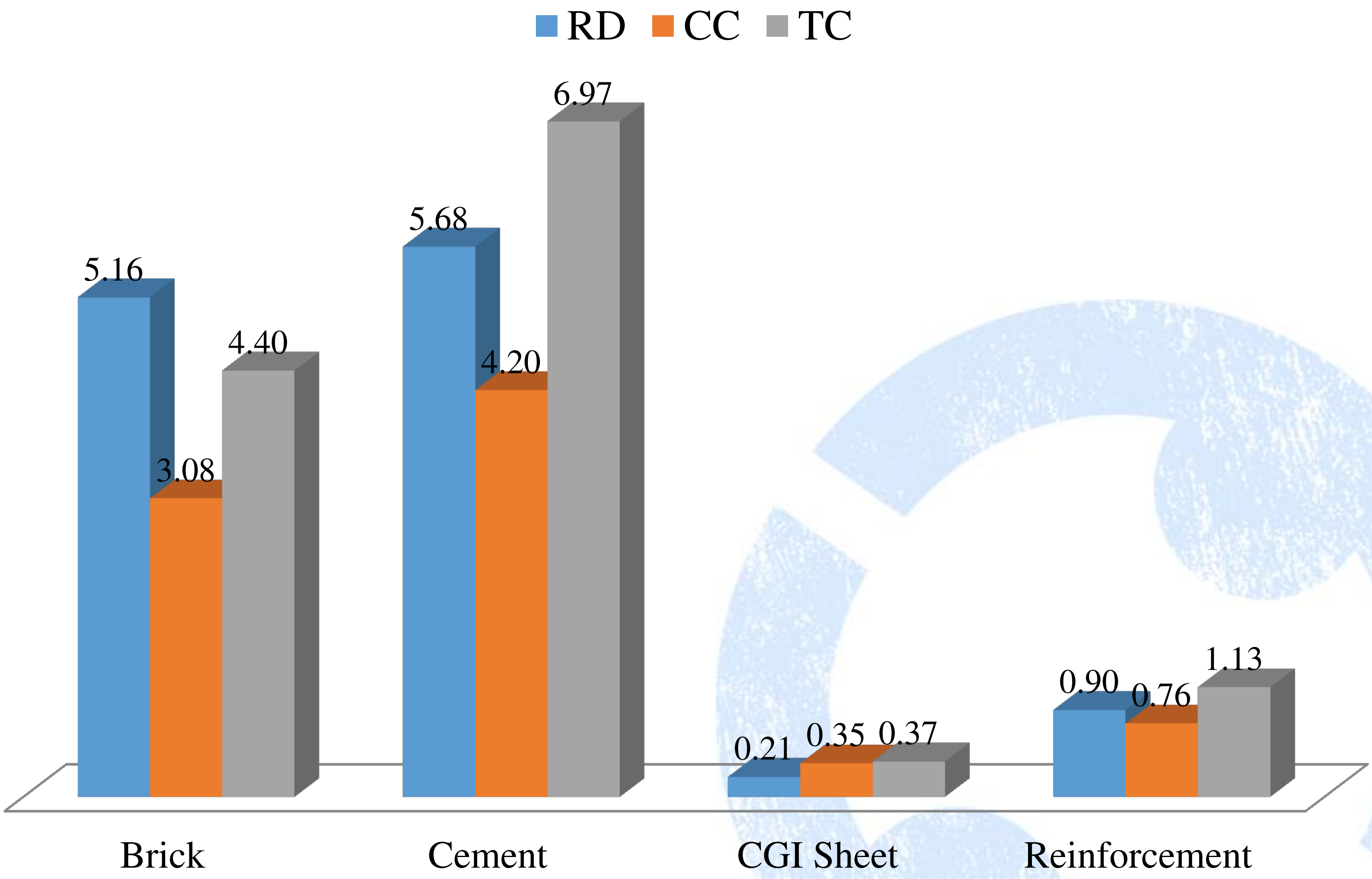
Year 1



RD: Regular annual demand + reconstruction demand
CC : Current production
TC : Install capacity

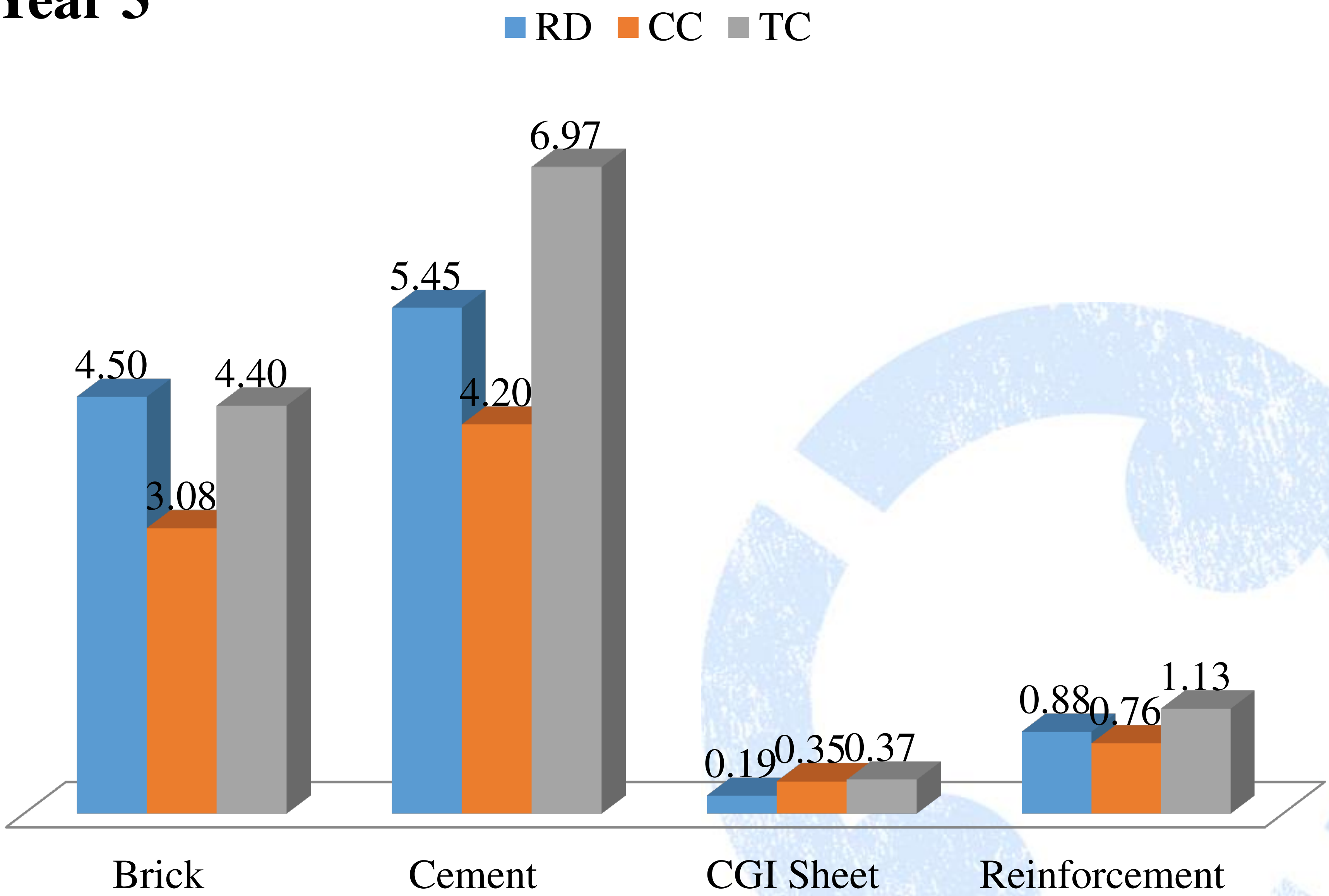
A. Demand and supply Gap

Year 2



A. Demand and supply Gap

Year 3



A. Demand and supply Gap

Major Issues

- **Legal**
- Forest act 1993 & Forest Regulation 1995 :
- National Parks Wildlife Conservation Act, 1973 (Section 5):
- Mines and Minerals Regulation act 1957
- Environmental Protection Regulation- 1998:
- Certification provision-1995
- Relaxation on the import tax on CGI sheet but no relaxation on raw materials
- Policy on import of CGI Sheet

Operational

- Power shortage
- Lack of skilled manpower
- Fuel Shortage
- Technology
- Frequent Strikes and Bandha

B) Price

Materials		PRICE					
		Factory	Regional	District	Depo	VDC (Semi Urban)	VDC (Rural)
Name	Unit	MRP	MRP	MRP	MRP	MRP	MRP
Cement	Bags	630	670	715		810	890
Brick	PCS	14	NA	NA	18	NA	NA
CGI	Bundle (50 KG)	5400	5650	5850	6090	6000	6200
Steel Bars	KG	62.5	68	70	73	75	80

B) Price

Issues

1. High production cost

Power shortage

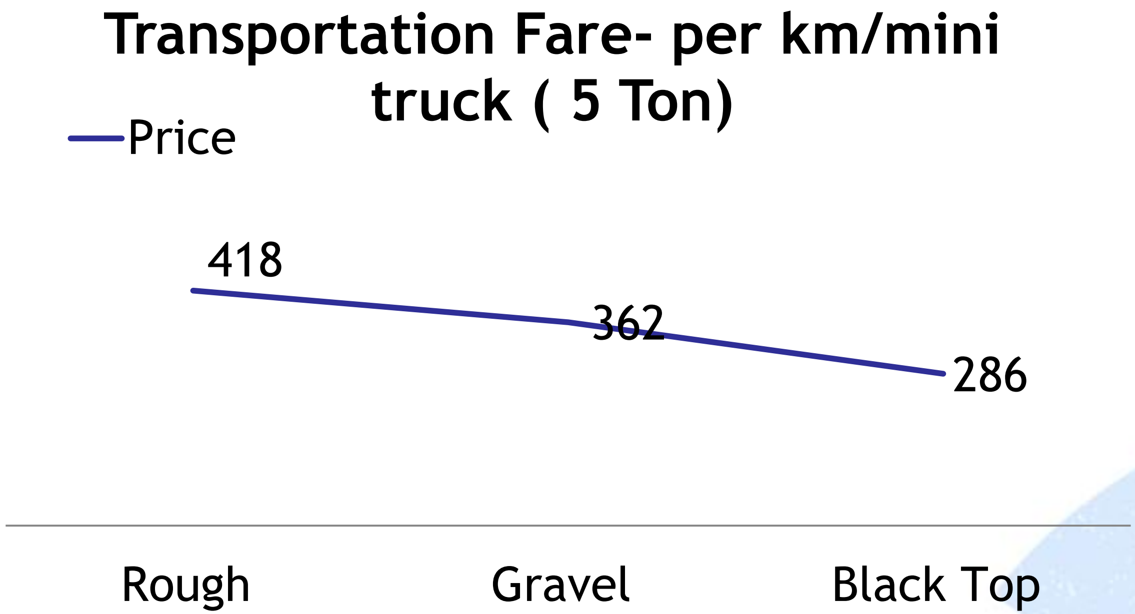
Price of raw materials

Labor shortage

Fuel Shortage

Periodic Interruption in production due to strikes and Bandha

2. High Transaction cost – transportation



The fare increase as high as Rs 125 /Km

Rs 600 to 900 for Porters for 40 kg

Difficulties	Percentage
Road condition	84.6
Unnecessary administrative interference	61.5
Absence of transportation groups	46.2
Parking Problem	15.4
Maintenance Problem	15.4

Issues

- Syndicate
- Hording of Materials

C. Quality

- CGI – Thickness/weight not up to the standard
- Cements – too many brands to choose from
- Bricks – Strength not up to the standard
- Blocks – not right mix, strength not up to the standard
- Sand- High silica context, Size not appropriate
- Aggregate : Pebbles, proper mix not maintained properly
- Stones – Random Rubbles, Corner /Through stone
- Timber- Not Seasoned and treated

Issues

- Awareness
- Technology
- No monitoring Mechanism
- Absence of testing facilities

Timely Supply

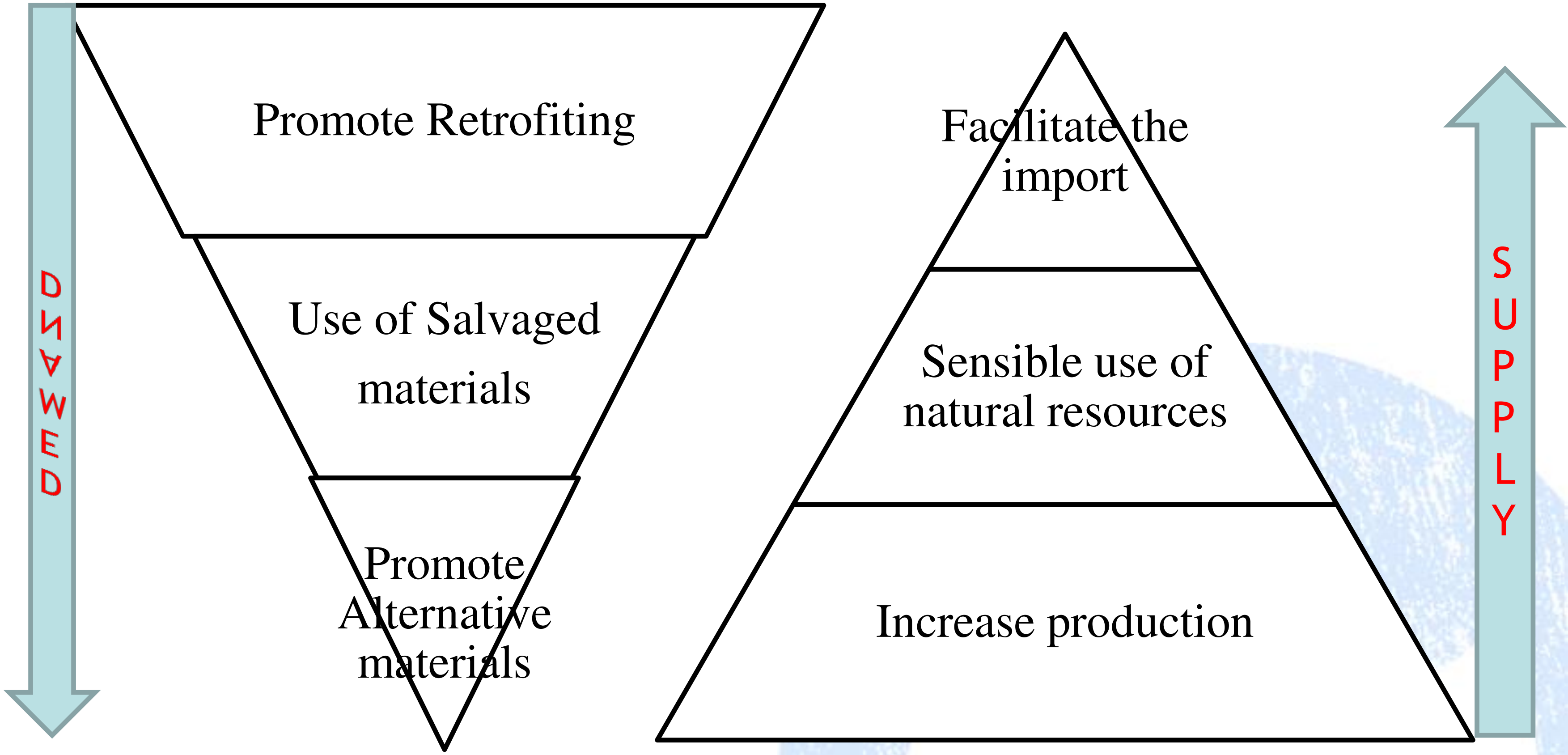
- No local materials – 2 weeks on an average
- Local materials : Several weeks sometimes

Timely Supply

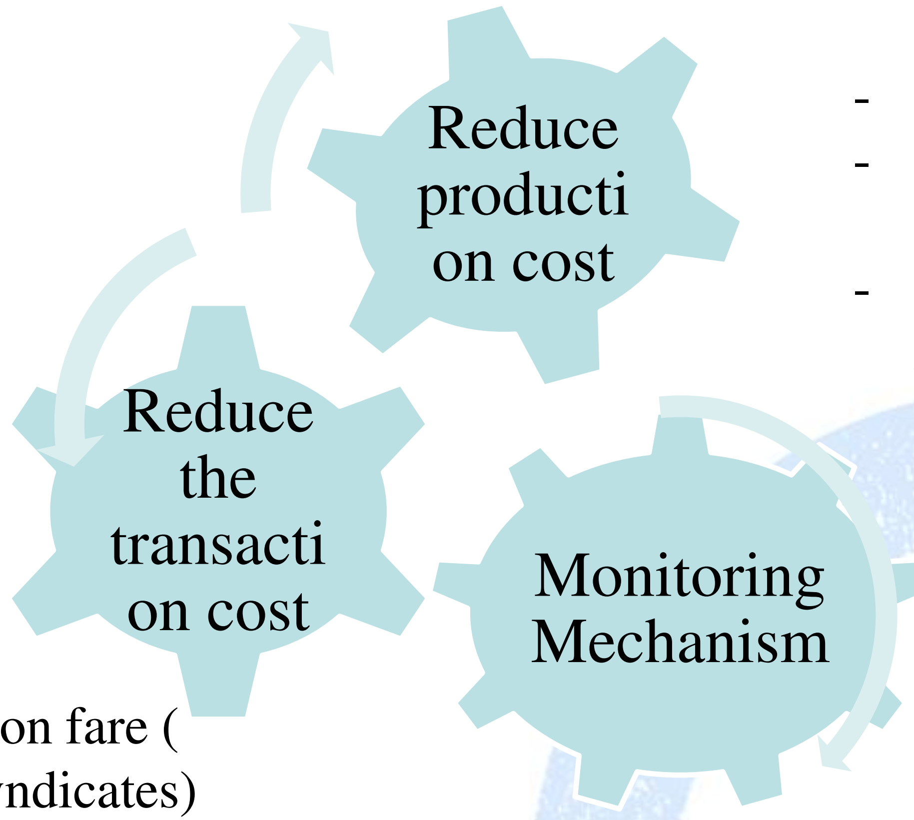
- Remoteness /bad road condition
- Limited local vendors /capacity- Storage Capacity
- Fuel Shortage
- Strikes
- Materials Hoarding

Strategy

Demand and Supply gap



Strategy
Price



- Power Supply
- Raw Materials (tax and duty wavier)
- Produce the skilled human resource at the local level

- Aggregation
- Fix transportation fare (dismantle the syndicates)
- Dealership at local level
- Subsidy in transportation for rural areas

- MIS System (Resource centers)
- Mobilize the state/parastal institutions to set price standard

Strategy Quality

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- Monitoring mechanism at district and national level
- Testing facilities at the district level

Technology

- Advanced Brick Kilns
- Stone cutting technology /Crushers
- Seasoning /treatments

Monitoring

Quality

Skill

- IEC materials including simple field testing techniques – single forum for IEC preparation
- Resource center
- Information desk at VDC level

Awareness

- Include materials component in mason training
- Trainings/orientations to producers

Strategy
Timely Supply

- Improve the transport infrastructure
- Strengthen the capacity of the local vendors
- Ensure availability of fuel in EQ affected districts
- Control hording of materials
- Relax rules around road tax/permits

Responsibilities

Government

- Review the policies that affect the production and supply of materials
- Set and monitor the quality and price standard, and enforce (*mobilize state/parastate institutions*)
- Improve transport infrastructures (*mobilize DoLIDAR*)
- Mobilize state/parastatal institutions to deliver the materials to remote areas (*Where private sector are unwilling to go*)
- Declare the industrial areas and the earthquake affected districts peace zone (no strikes, Bandhs)
- Demand aggregation
- Set up a coordination committee to oversee the supply chian

Responsibilities

<u>Private Sector</u>

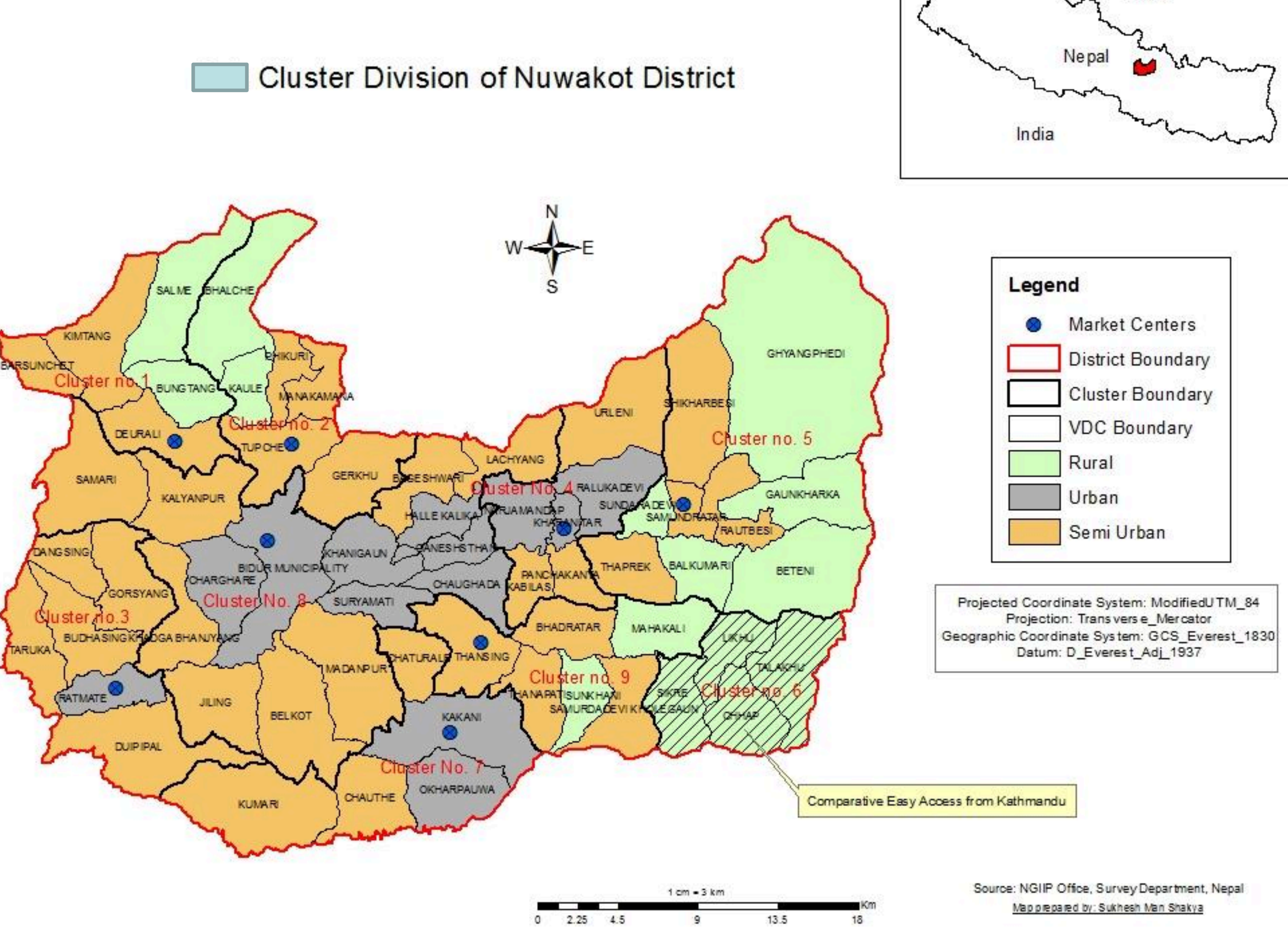
- Lead the supply of the construction materials (as they already have functioning network)

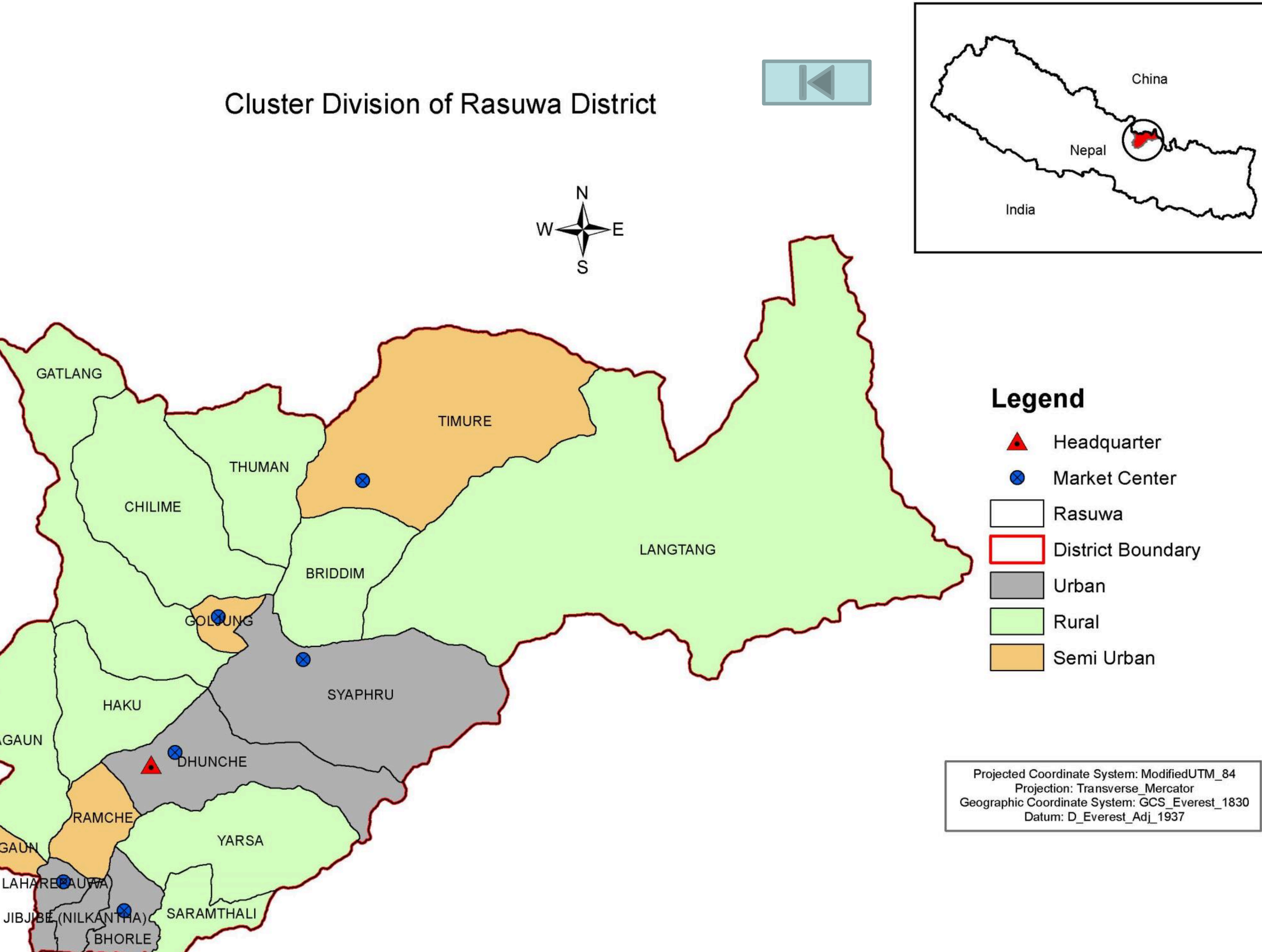
<u>Development Partners</u>

- Study/Research
- Awareness- develop ICE materials, MIS system
- Capacity building of producers and vendors
- Technology adaptation (mainly on the alternative construction materials)

Conclusion

There are gaps in all the dimensions (*Supply and demand, quality, price and timely supply*) of the supply chain. The gap is more in local materials. While largely, the operational level interventions can address issues around the non local materials, both operational and policy level interventions are required to improve the supply of local materials.


Cluster Division of Nuwakot District
Nepal
India
N
S
E
W
Legend
Market Centers
District Boundary
Cluster Boundary
VDC Boundary
Rural
Urban
Semi Urban
Projected Coordinate System: ModifiedUTM_84
Projection: Transverse_Mercator
Geographic Coordinate System: GCS_Everest_1830
Datum: D_Everest_Adj_1937
Comparative Easy Access from Kathmandu
Cluster no.1
Cluster no. 2
Cluster no.3
Cluster No. 4
Cluster no. 5
Cluster no. 6
Cluster No. 7
Cluster No. 8
Cluster no. 9
SALME
BHALCHE
KIMTANG
BARSUNCHET
BHIKURI
BUNGTANG
KAULE
MANAKAMANA
DEURALI
TUPCHE
SAMARI
KALYANPUR
GERKHU
BAGESHWARI
LACHYANG
URLENI
SHIKHARBESI
GHYANGPHEDI
RALUKADEVI
SUNDARADEVI
SAMUNDRATAR
GAUNKHARKA
RAUTBESI
HALLE KALIKA
KHADGABHANJYANG
KHANIGAUN
GANESHSTHAN
BIDUR MUNICIPALITY
CHARGHARE
DANGSING
GORSYANG
BUDHASING
TARUKA
SURYAMATI
CHAUGHADA
KABILAS
PANCHAKANYA
THAPREK
BALKUMARI
BETENI
BHADRATAR
MAHAKALI
LIKHU
TALAKHU
MADANPUR
CHATURALE
THANSING
RATMATE
JILING
BELKOT
KAKANI
THANAPATI
SUNKHANI
SAMUNDRADEVI
KHOLEGAUN
SIKRE
CHHAP
DUIPIPAL
KUMARI
CHAUTHE
OKHARPAUWA
1 cm = 3 km
0
2.25
4.5
9
13.5
18
Km
Source: NGIIP Office, Survey Department, Nepal
Map prepared by: Sukhesh Man Shakya


Cluster Division of Rasuwa District
N
W
E
S
China
Nepal
India
GATLANG
TIMURE
THUMAN
CHILIME
BRIDDIM
LANGTANG
GOLJUNG
SYAPHRU
HAKU
DHUNCHE
RAMCHE
YARSA
LAHARE PAUWA
JIBJIBE (NILKANTHA)
SARAMTHALI
BHORLE
Legend
Headquarter
Market Center
Rasuwa
District Boundary
Urban
Rural
Semi Urban
Projected Coordinate System: ModifiedUTM_84
Projection: Transverse_Mercator
Geographic Coordinate System: GCS_Everest_1830
Datum: D_Everest_Adj_1937

Demand

Types of houses before earth quake and after (Intended)										
	Mud bonded stone		Mud bonded brick		Cement bonded stone		Cement bonded brick		Other	
Nuwakot	Before	After	Before	After	Before	After	Before	After	Before	After
Urban type	76.1	29.1	2.5	2.6	2.5	11.9	16.5	51.6	2.5	4.7
Semi-urban type	89.4	57	1.9	2.2	1.7	13.6	6.1	22.2	0.9	5
Rural type	97.8	79.2	0	0.4	0.4	6.9	1.6	9.8	0.2	3.6
Total	**87.9**	**55.2**	**1.6**	**1.9**	**1.6**	**11.8**	**7.8**	**26.4**	**1.1**	**4.7**
Rasuwa										
Urban type	72.8	55	0	0	7.2	14.9	5	11.8	15	18.3
Semi-urban type	63.9	35.2	2.7	0.9	11.4	39.2	3.4	13.7	18.6	11
Rural type	72.9	60.2	0	0	2.2	16.7	2.3	5.2	22.6	17.8
Total	**69.9**	**50.6**	**0.9**	**0.3**	**6.6**	**23.6**	**3.4**	**9.9**	**19.1**	**15.7**

Thank you

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